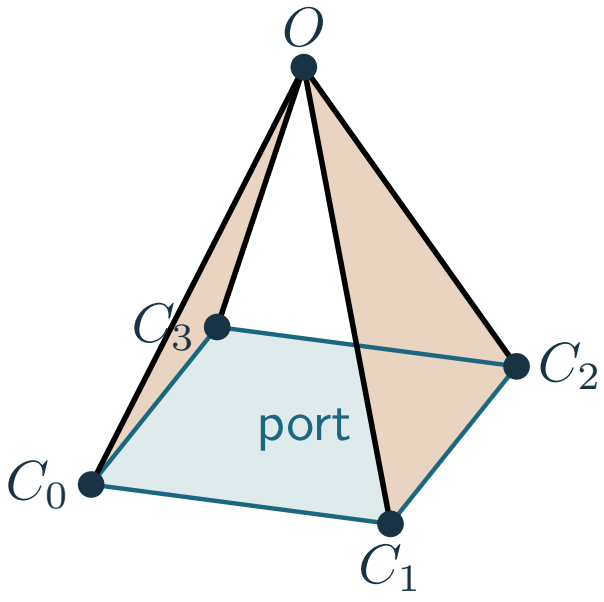
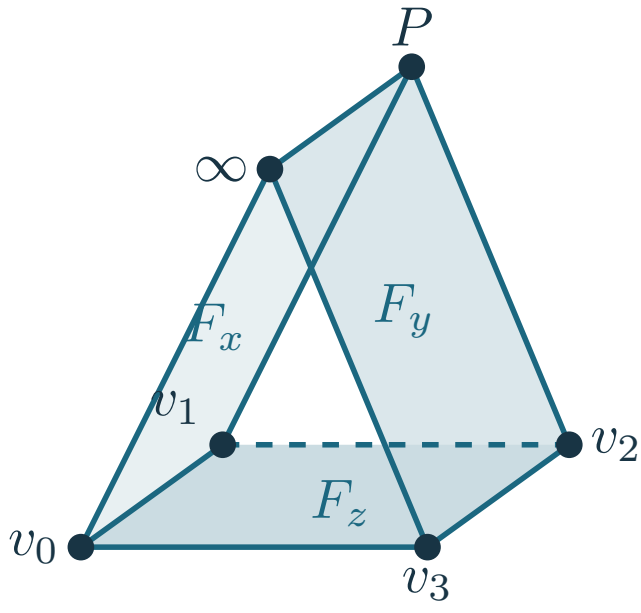
Complete-circle leaf



$\mathrm{Vol} = \pi g(x)$

Two shaded triangles retain the actual crossing-disk pairing.

Multiplication cell



$\mathrm{Vol} = \pi m(x, y, z)$

Three ordered quadrilateral ports; two triangular projection faces.

Stationarity matches the port cross-ratios: $\kappa_{\mathrm{child}}\kappa_{\mathrm{parent}} = 1$.
Combinatorial projections: every vertex is ideal.